Relevance of AI to major Engineering Fields

1. Electrical / Electronics Engineering

•**Predictive Maintenance-** AI algorithms analyze data from sensors on electrical equipment (like motors, generators, and transformers) to predict when maintenance is needed.

•**Intelligent Control Systems:** AI-powered control systems are being used in various applications, including robotics, industrial automation, and renewable energy systems.

- •For example, AI can optimize the performance of wind turbines by adjusting blade angles for maximum energy capture.

•**Power Quality Improvement**

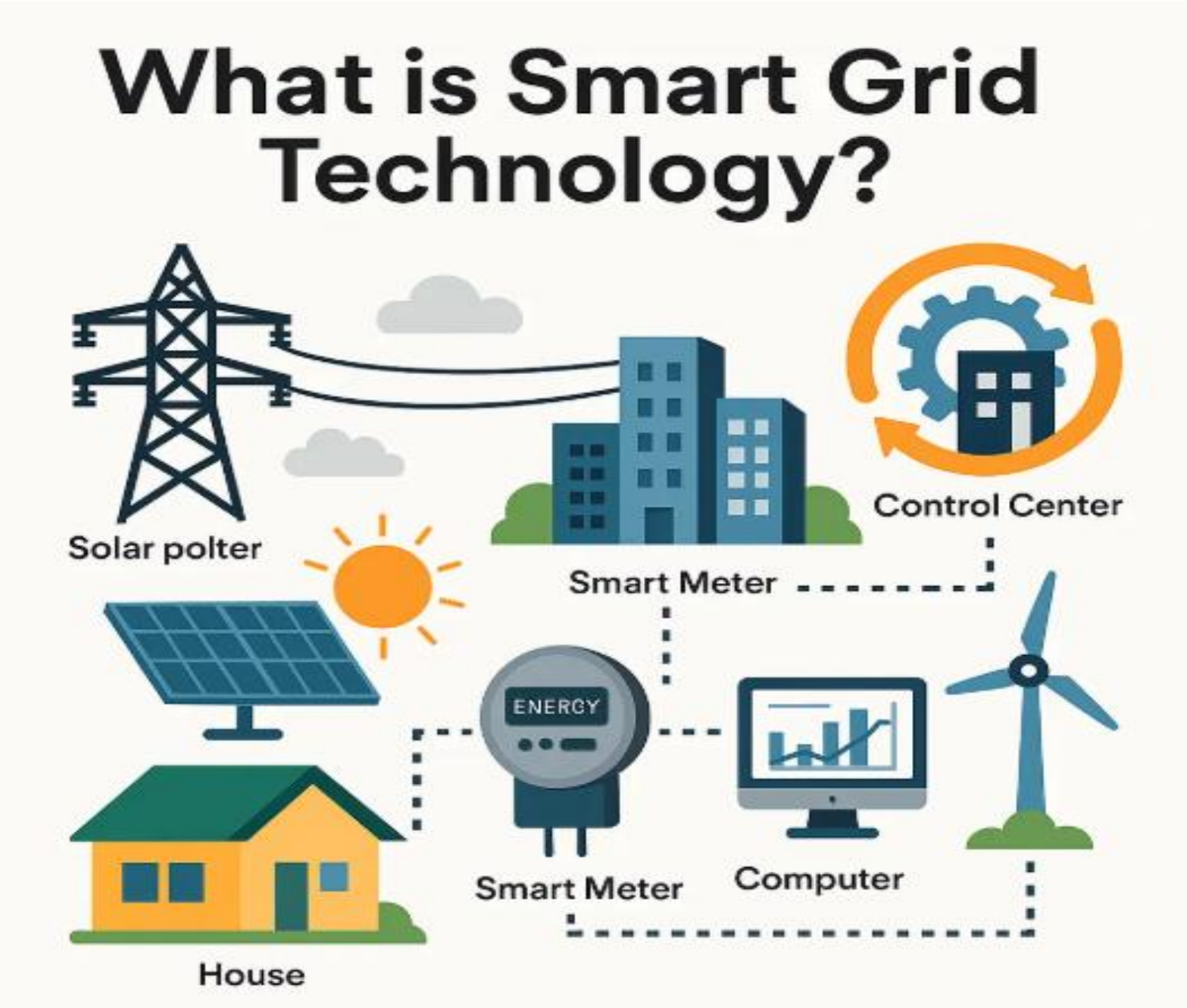
•**AI in Electric Vehicles**

Electrical Engineering - Smart grid, power system automation

•**Example**: Power Quality Improvement

Steps:

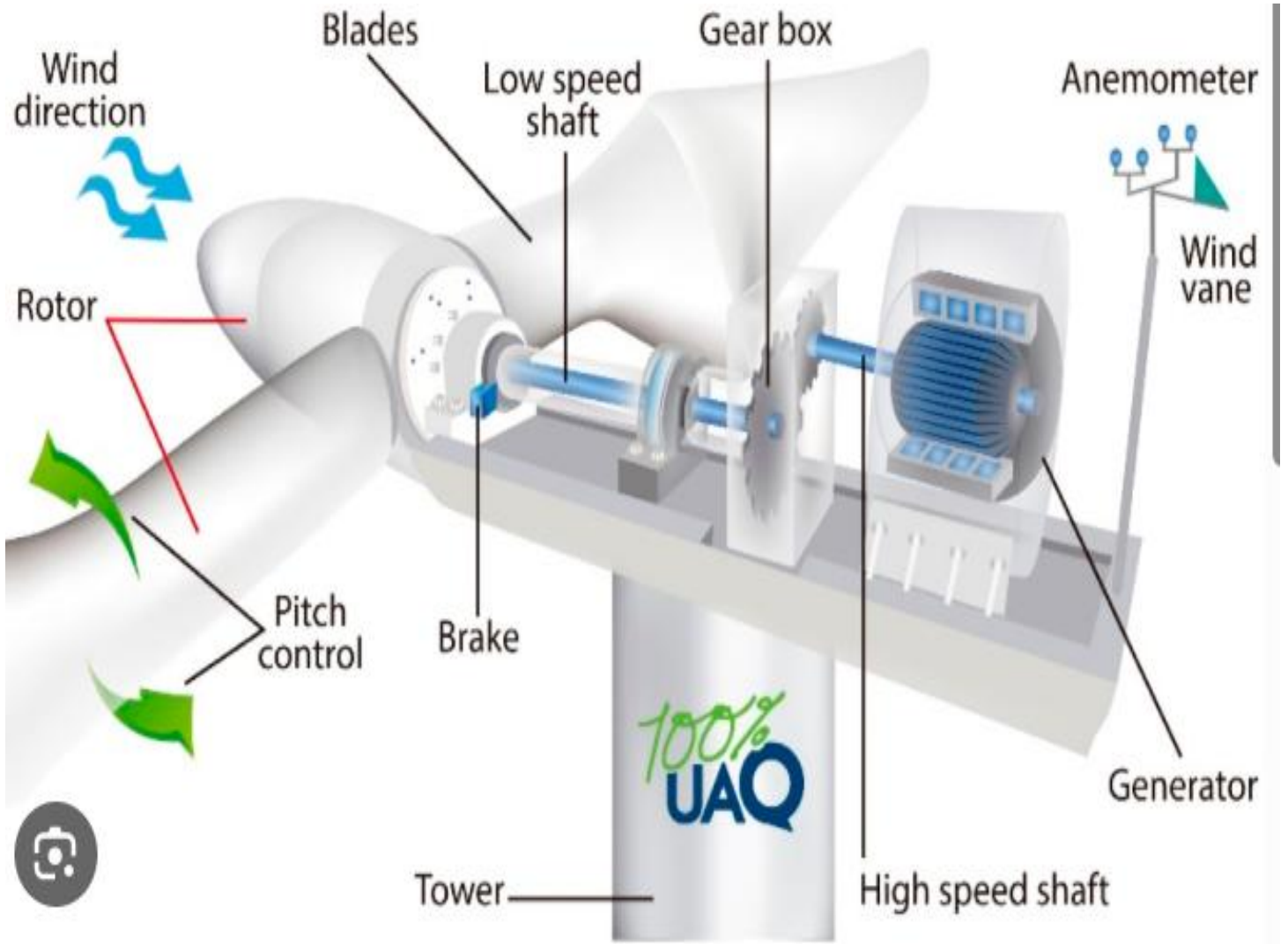
1. Detect voltage/frequency faults via sensors
2. Input to neural network/expert system
3. Diagnose fault type
4. Trigger corrective action
5. Improve stability and reduce outages

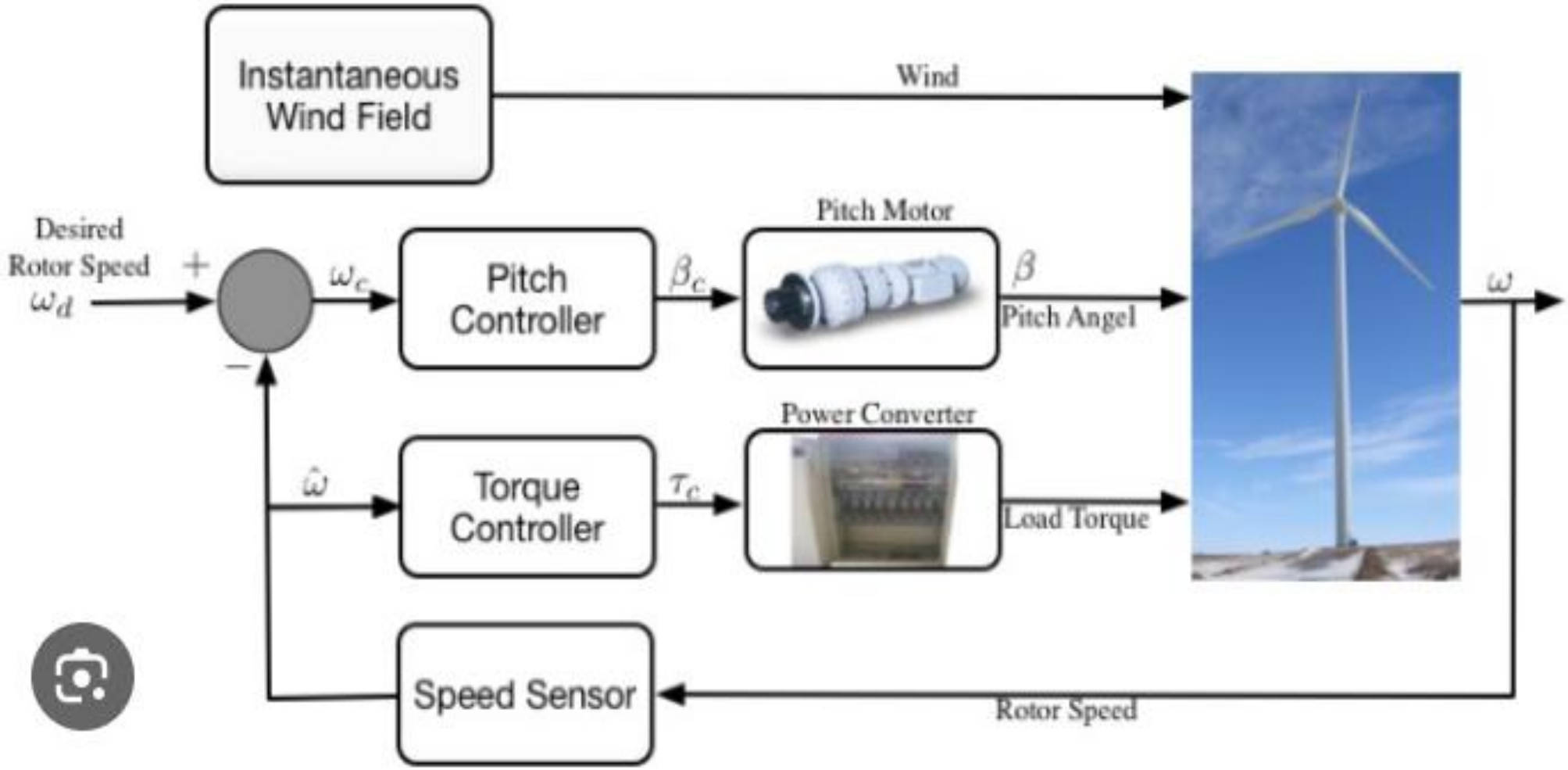


The smart grid is an innovative power system based on information and communication technology. It realizes real-time monitoring, intelligent scheduling, and optimal management of the power system through smart, communication networks and computer technology sensors

- **Real-time Power Output Optimization:** Wind conditions are anything but constant. AI systems continuously monitor weather patterns and adjust turbine settings in real-time to maximize energy capture.

- By fine-tuning factors like blade angle and rotation speed, AI ensures turbines operate at optimal efficiency. Think of it as a smart autopilot for wind farms, always adjusting to harness the most power from every breeze.





AI can optimize the performance of wind turbines by adjusting blade angles for maximum energy capture

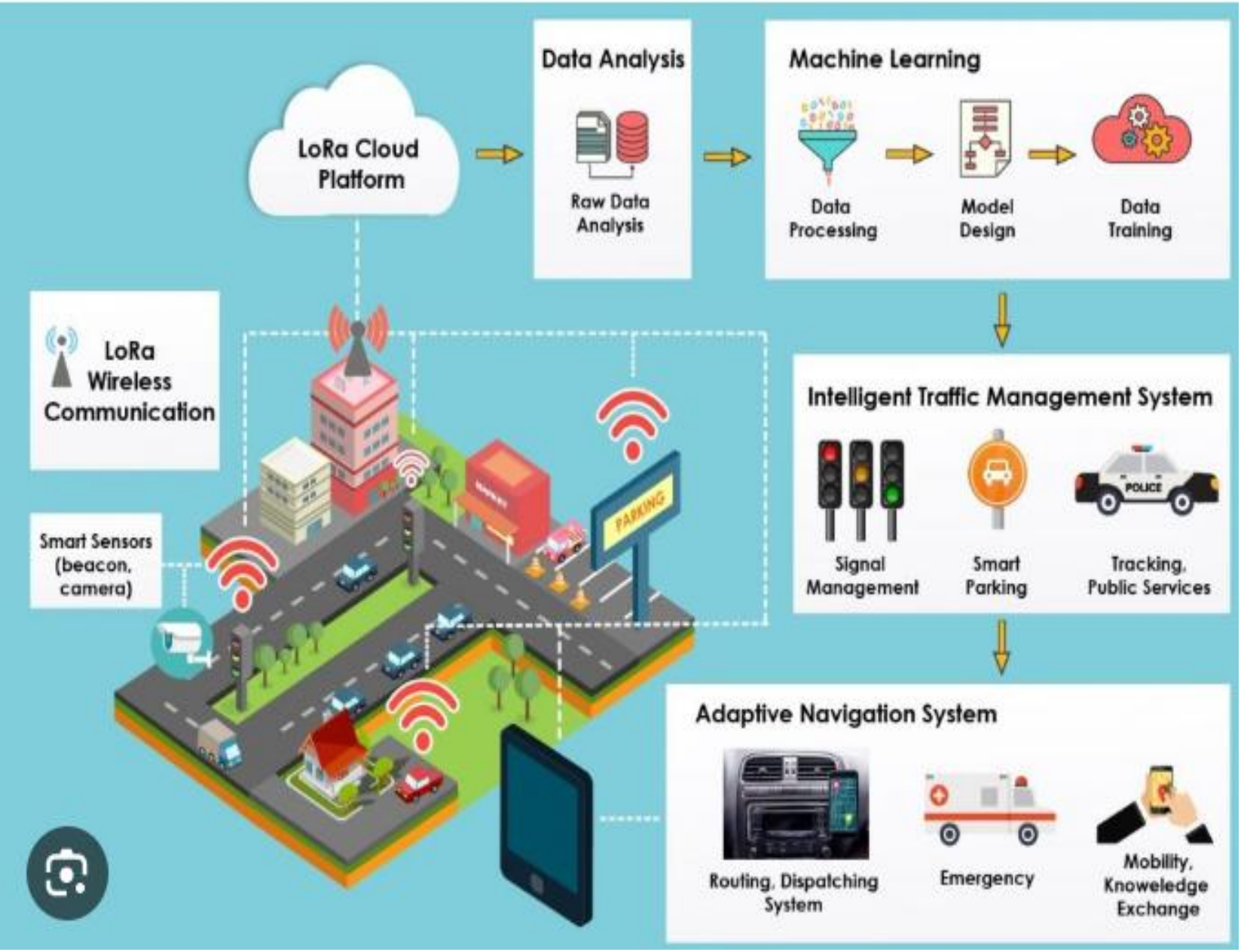
2. Civil Engineering

•Traffic management, construction planning –

Example: Smart Traffic Signals –

•**Steps:**

1. Live traffic data from cameras
2. AI processes vehicle flow
3. Adjusts signal timing dynamically
4. Reduces congestion
5. Learns traffic trends



Smart traffic systems use data from sensors and connected infrastructure to adjust signal timing and traffic flow.

By responding to real world conditions, such as congestion or vehicle volume, these systems help reduce delays and improve consistency for public transit and commuters

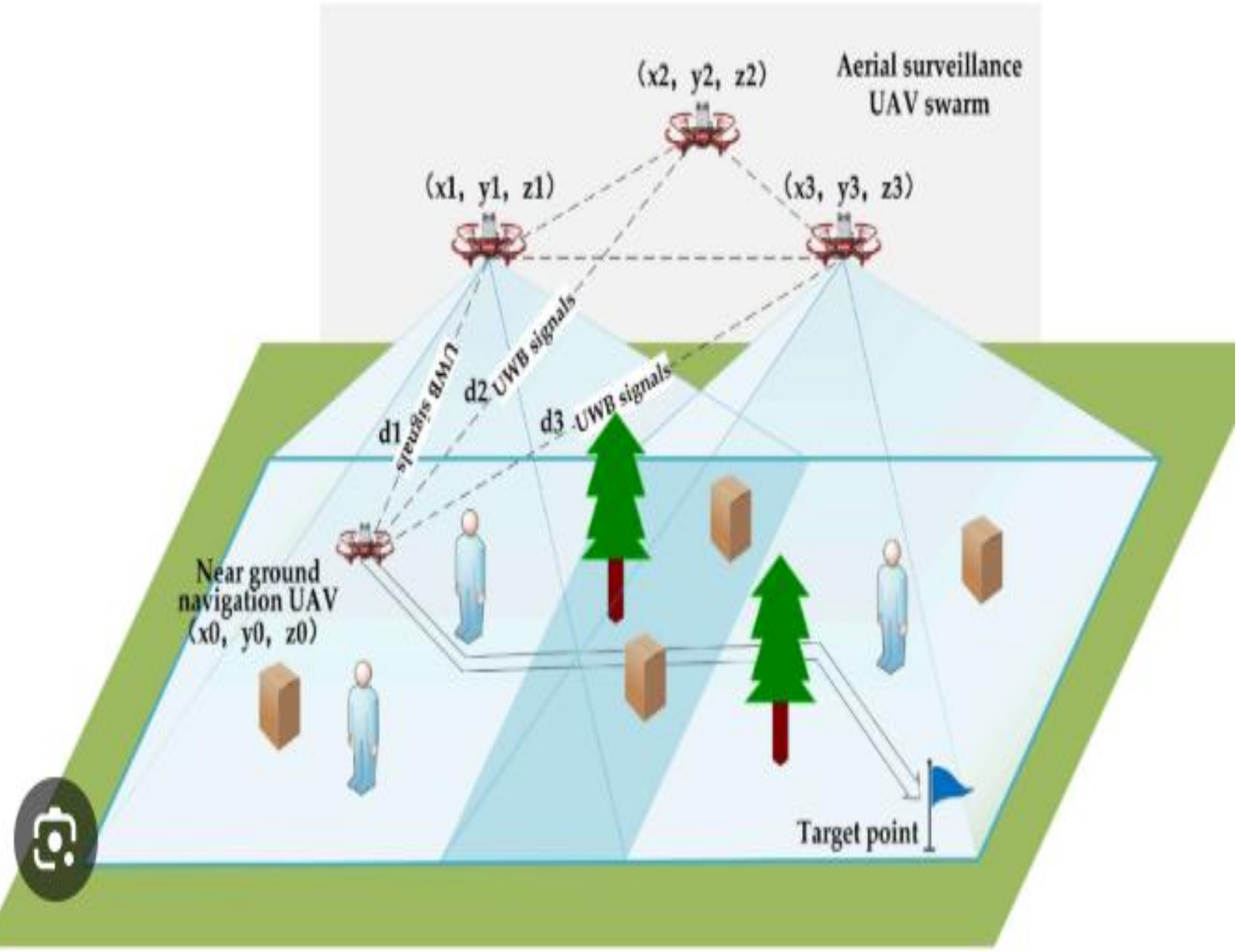
3. Aerospace Engineering

Relevance: Autonomous navigation, flight safety

Example: Drone Navigation

Steps:

1. Sensors detect environment
2. AI processes GPS and visual data
3. Plans safest flight route
4. Avoids obstacles



- Drone navigation refers to the ability of a drone to move in the air and determine its position while following a specific destination or route.

- This includes the use of GPS, GNSS and other sensors as well as the application of navigation algorithms to calculate and control a precise flight path.

4. Mechanical Engineering

- **Design Optimization**
- **Automation and Robotics**
- **Maintenance prediction, automation**

Example: Predictive Maintenance in Factories –

•**Steps**:

1. Sensors collect vibration/temperature data
2. AI model analyzes for failure patterns
3. Alert maintenance team
4. Fix before failure occurs
5. Increase machine life



- Predictive maintenance in manufacturing is revolutionizing the manufacturing industry, offering proactive strategies to optimize operations.

- By leveraging AI, machine learning, and data analytics, companies can reduce downtime, increase efficiency, and cut operational costs

5. Industrial Engineering

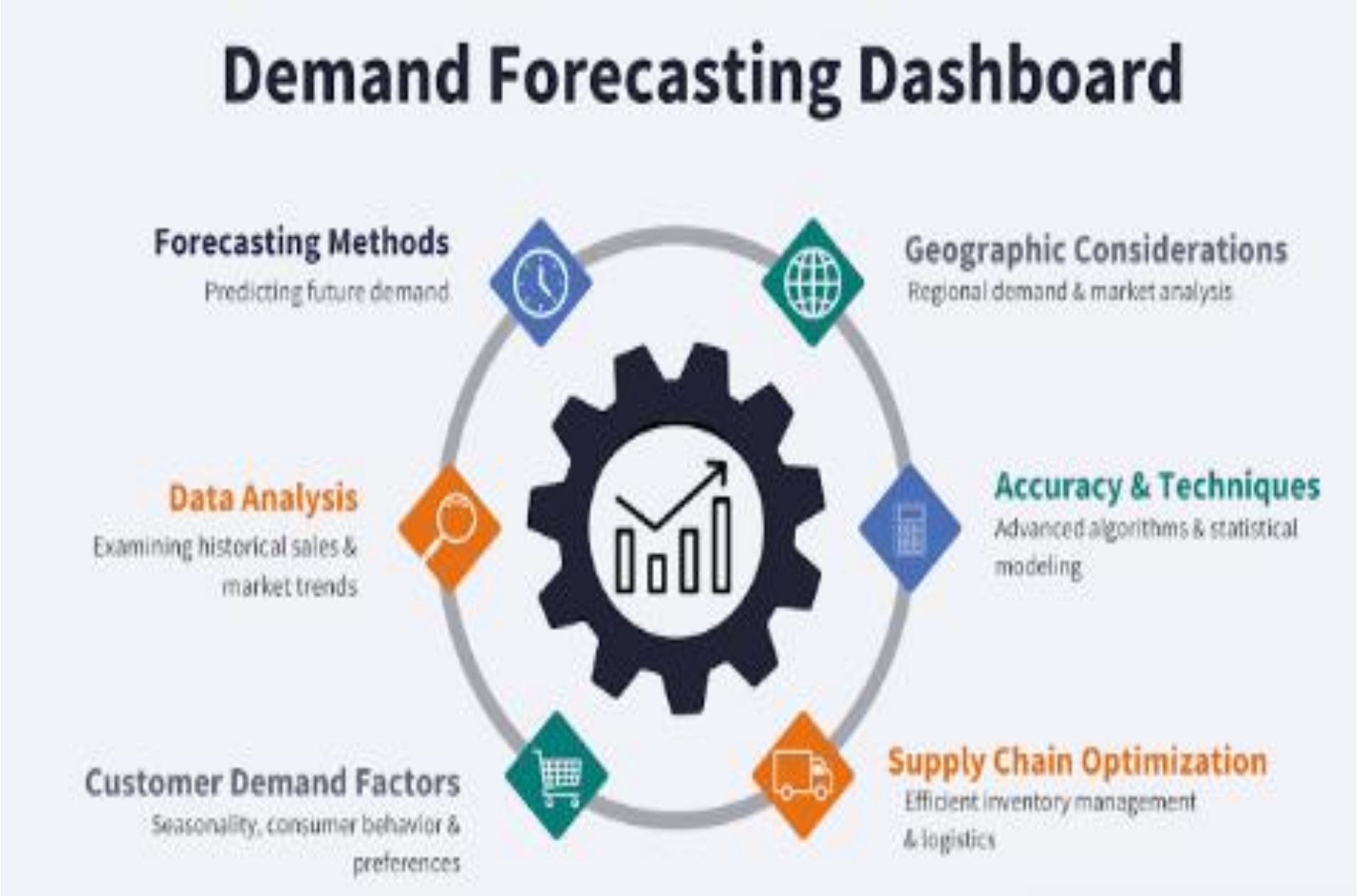
Supply chain, inventory, production –

- **Example**: Demand Forecasting

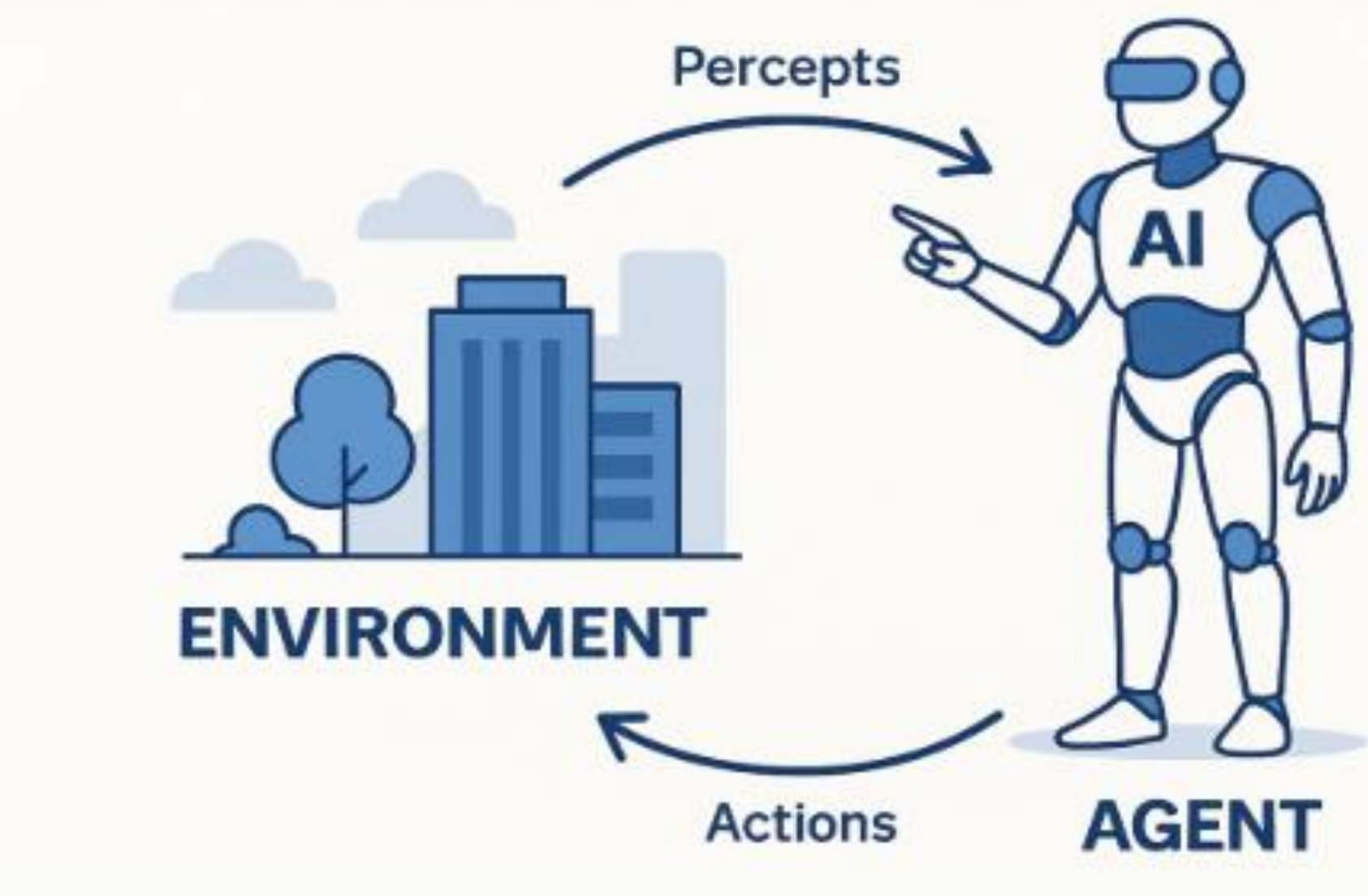
- **Steps**:

1. Collect sales data
2. Use AI to predict demand
3. Automate reordering
4. Avoid over/under stocking

- Demand forecasting is the process of predicting future demand for a product.
- It's about understanding how many units your customers will want to buy and when so you can make sure you have the right amount of inventory to meet their demands



AGENT AND ITS ENVIRONMENT

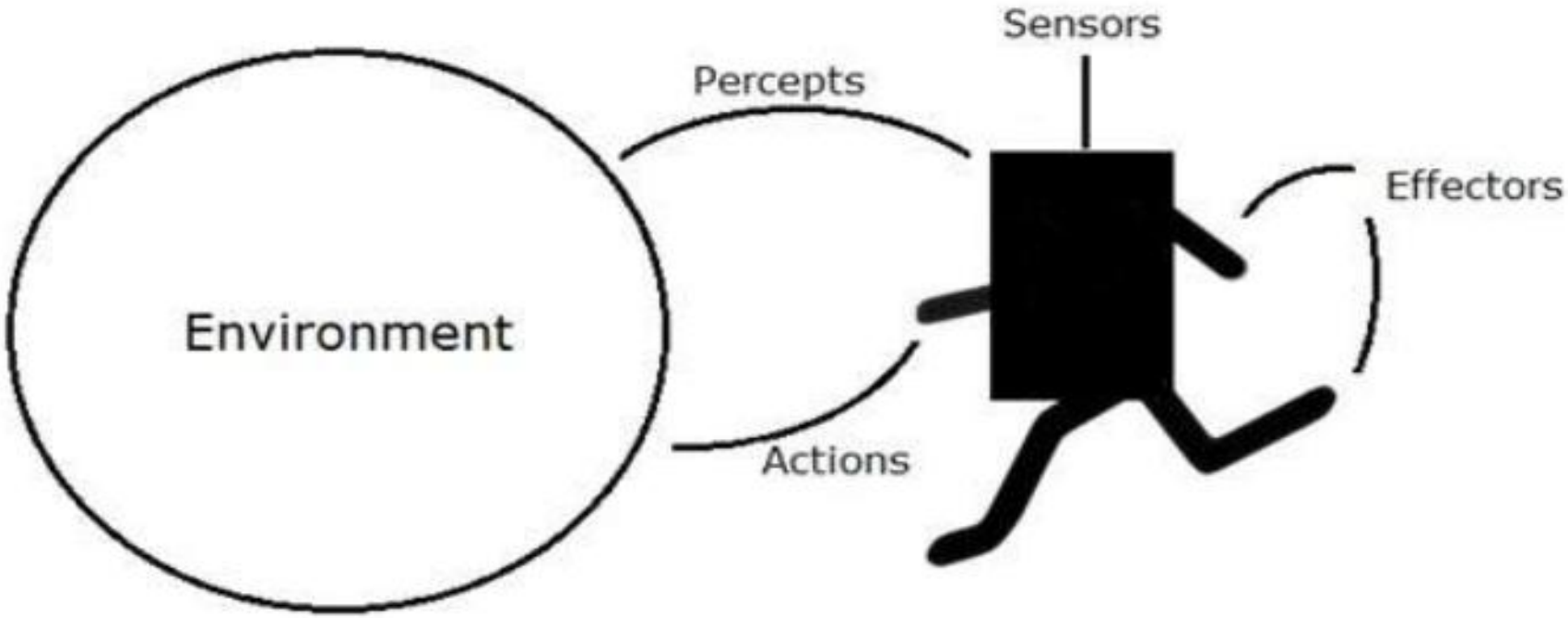


- A **human agent** has sensory organs such as eyes, ears, nose, tongue and skin parallel to the sensors, and other organs such as hands, legs, mouth, for effectors.

- A **robotic agent** replaces cameras and infrared range finders for the sensors, and various motors and actuators for effectors.

- A **software agent** has encoded bit strings as its programs and actions.

- An agent is anything that can perceive its environment through its sensors and act upon that environment through its effectors.



- In the context of Artificial Intelligence, an agent is a system that can perceive its surroundings, make decisions, and take actions to achieve its specific goals.

Agent Terminology -

- **Performance Measure of Agent** − It is the criteria, which determines how successful an agent is.
- **Behaviour of Agent** − It is the action that agent performs after any given sequence of percepts.
- **Percept** − It is agents perceptual inputs at a given instance.
- **Percept Sequence** − It is the history of all that an agent has perceived till date.
- **Agent Function** − It is a map from the precept sequence to an action. It is an abstract mathematical description.
- **Agent Program** is a concrete implementation, running within some physical system.

Characteristics of Agents:

- **Autonomy:** The ability to operate independently and make decisions without constant human oversight.

- **Reactivity:** The capability to respond to environmental stimuli and changes.

- **Proactivity:** The ability to exhibit goal-directed behaviour, anticipate future states, and plan actions to achieve long-term objectives.

- **Social Ability:** For multi-agent systems, this involves the capacity to interact, coordinate, or compete with other agents to achieve goals.

- **Rationality:** An agent is rational if it chooses actions that maximize its expected outcome or performance.

- **Adaptability/Learning:** Some agents can learn from past experiences and refine their behavior to improve performance over time

RATIONAL AGENT

- *A **rational agent** is one that does the right thing—conceptually speaking, every entry in the table for the agent function is filled out correctly.*

Rationality of an agent depends on the following –

- The **performance measures**, which determine the degree of success.
- Agents **Percept Sequence** till now.
- The agents **prior knowledge about the environment**.
- The **actions** that the agent can carry out.

What is Ideal Rational Agent?

- An ideal rational agent is the one, which is **capable of doing expected actions to maximize its performance measure**, on the basis of −

    - Its percept sequence
    - Its built-in knowledge base

SPECIFYING THE TASK ENVIRONMENT

- **P - Performance measures**
- **E - Environment**
- **A - Actuators**
- **S - Sensors**

- **Performance measures:** These are the parameters used to measure the performance of the agent. How well the agent is carrying out a particular assigned task.

- **Environment:** It is the task environment of the agent. The agent interacts with its environment. It takes perceptual input from the environment and acts on the environment using actuators.

- **Actuators:** These are the means of performing calculated actions on the environment. For a human agent; hands and legs are the actuators.

- **Sensors:** These are the means of taking the input from the environment. For a human agent; ears, eyes, and nose are the sensors.

Let us come back to the auction activity.

Performance measures:	**Environment:**	**Actuators: (means to perform the activity)**	**Sensors: (means to perceive the environment)**
Safe, Fast Legal, Comfortable trip Maximise trip	Road, traffic, pedestrians, customers	Steering, Accelerator, brake, signal, horn, display	Cameras, sonar, speedometer, GPS, odometer, accelerometer, engine sensors, keyboard

Let us come back to the auction activity.

Performance measures:	**Environment:**	**Actuators: (means to perform the activity)**	**Sensors: (means to perceive the environment)**
1. Cost of the item 2. Quality of the item 3. Value of the item 4. Necessity of the item	1. Auctioneer 2. Bidders 3. BiddersItems which are to be bid	1. Speakers 2. Microphones 3. Display items 4. Budget	1. Camera 2. Price monitor, where prices are being displayed. 3. Eyes 4. Ears of the attendees

Agent	Performance Measure	Environment	Actuators	Sensors
Hospital Management System	Patient health, admission efficiency, payment processing	Hospital, doctors, patients	Prescriptions, diagnoses, scan reports	Symptoms, patient responses
Autonomous Car	Safety, comfort, travel efficiency	Roads, traffic, vehicles	Steering, accelerator, brakes	Camera, GPS, odometer
Intelligent Tutoring System	Student performance and learning improvement	Classroom, students, learning resources	Smart displays, feedback, corrections	Student responses, assignments
Part-Picking Robot	Percentage of parts placed correctly	Conveyor belt, parts, bins	Robotic arms and grippers	Camera, position sensors
Satellite Image Analysis System	Accuracy of image classification	Satellite image data	Display categorized results	Image pixel data

TYPES OF AGENT

- Simple reflex agents
- Model-based reflex agent
- Goal-based agents
- Utility Based agents

1. Simple reflex agents

Simple reflex agents act only on the **current perception** of the environment using predefined condition–action rules. They **do not rely on past experiences or predict future** outcomes and respond directly using simple "if–then" logic.

- Operate without memory of past states or stored information
- Do not maintain any internal model of the environment
- Work best in fully observable and predictable environments

For Example, Traffic light control systems that change signals based on fixed timing.

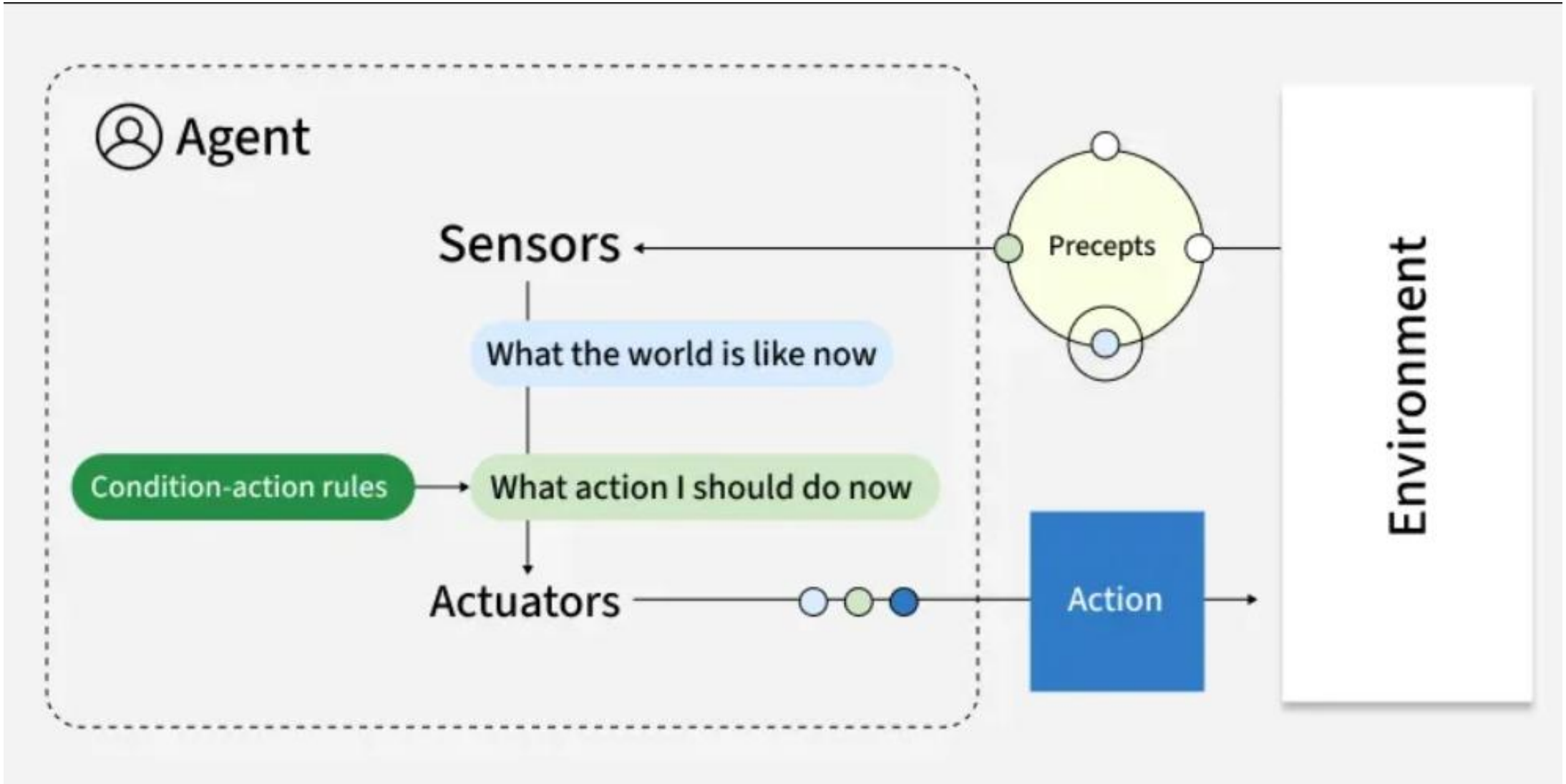


Figure: Simple reflex agents

2. Model-based reflex agent

Model-based reflex agents **maintain an internal model of the environment** to handle situations where full information is not directly available. This helps them make better decisions by considering changes in the environment and the impact of their actions.

- Use internal memory to update and maintain environment state
- Infer missing details from partial or incomplete observations
- React based on current input while using stored context for better decisions

    - For example, Robot vacuum cleaners that map rooms and tracks cleaned areas.

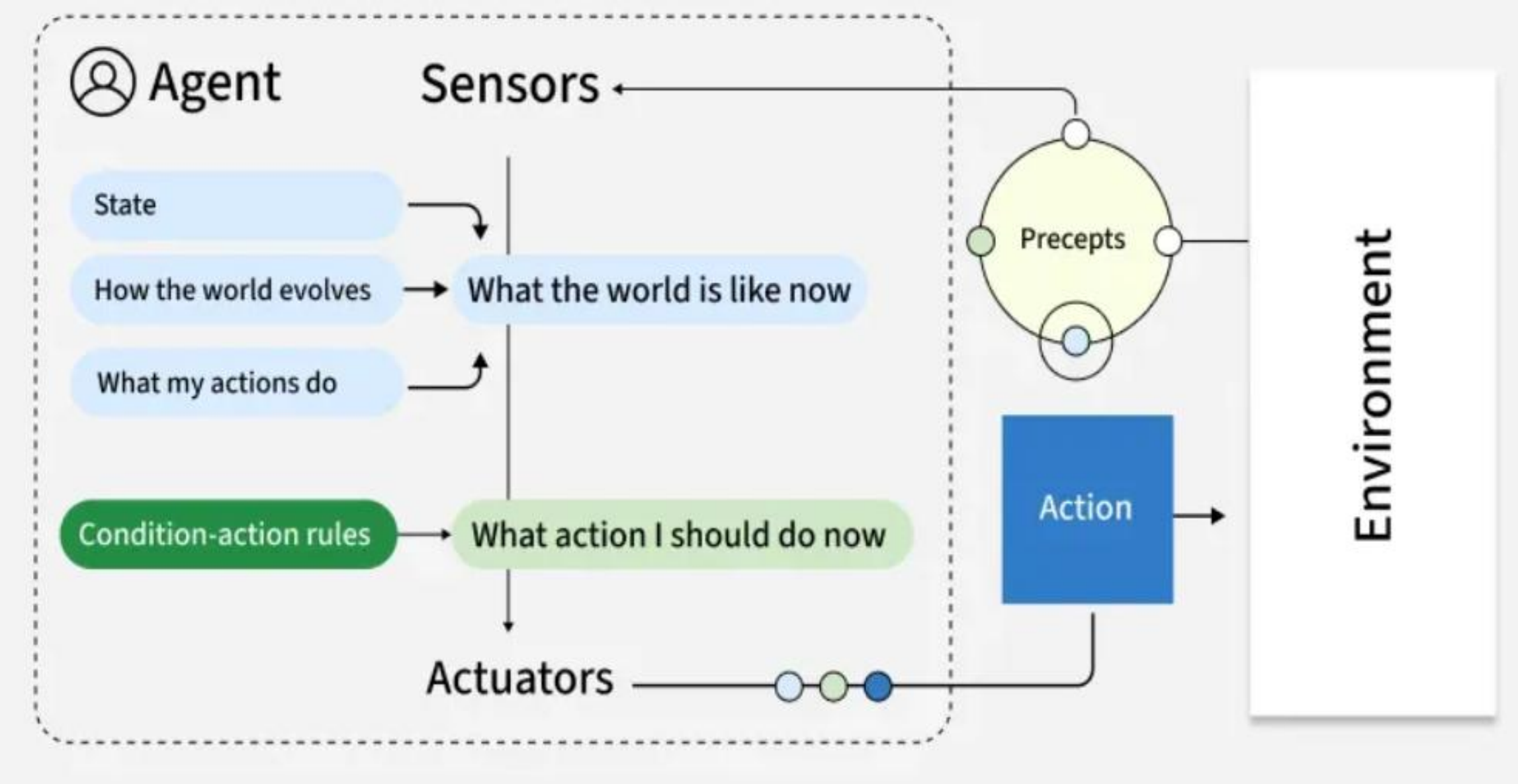


Figure: Model-based reflex agent

3. Goal-based agents

Goal-based agents choose their actions by focusing on a specific objective and evaluating how different choices can help achieve it. Instead of reacting only to the current situation, they plan ahead and consider possible future outcomes.

- Use search and planning techniques to find suitable action paths
- Judge actions based on how well they move toward the goal
- Consider future consequences before making decisions
- Explore multiple possible routes to reach the desired outcome

    - For example, Logistics routing agents that find optimal delivery routes based on factors like distance and time. They continually adjust to reach the most efficient route.

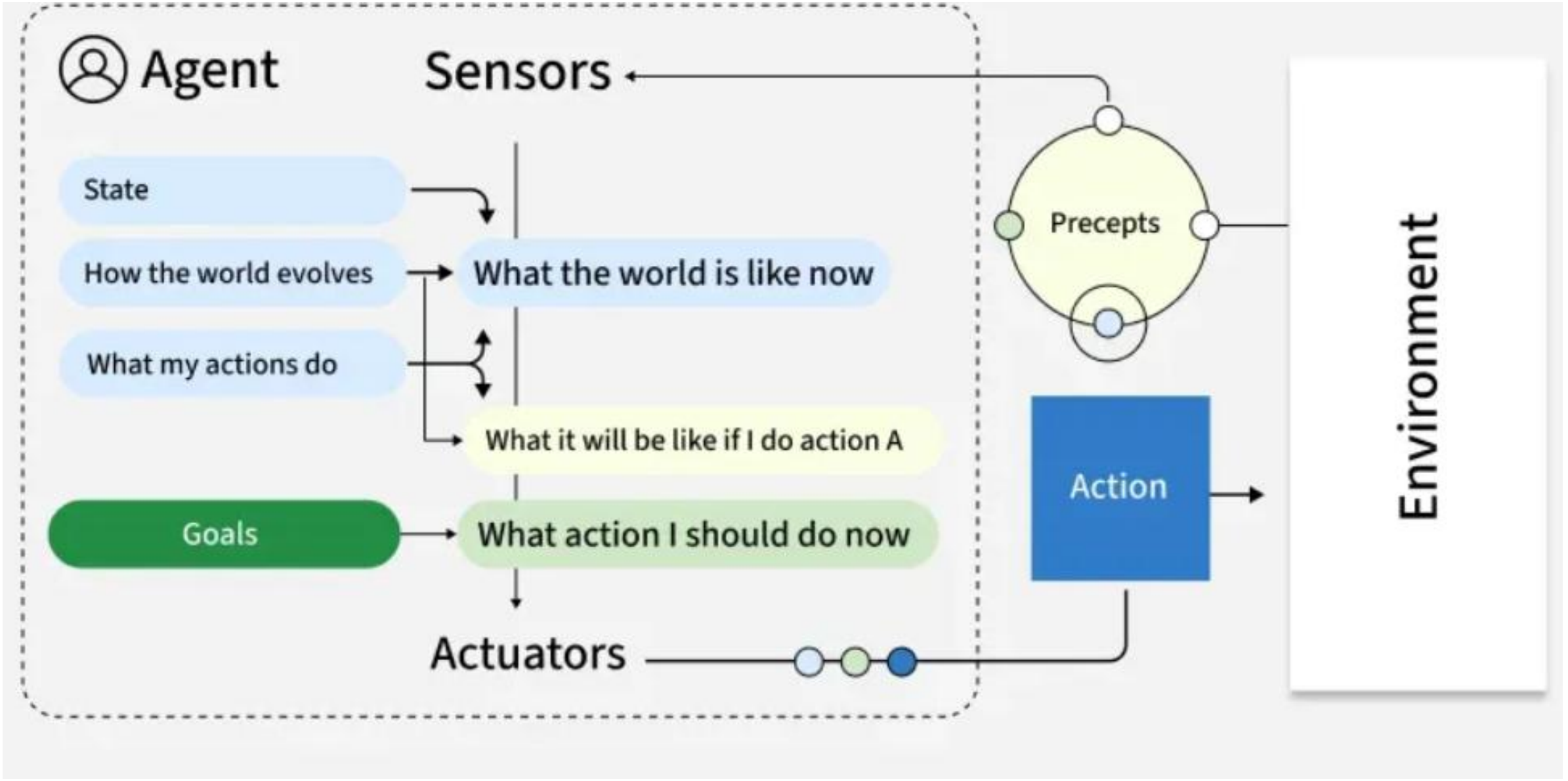


Figure: Goal-based agents

4. Utility-Based Agents

Utility-based agents go beyond simply achieving goals by evaluating how beneficial each action is using a utility function, which measures the overall "value" or satisfaction of an outcome. This helps them choose the best option when dealing with trade-offs or uncertainty.

- Make decisions in uncertain or probabilistic environments
- Select actions based on expected usefulness or reward
- Optimize choices while working within given constraints

  - For example, Financial portfolio management agents that evaluate investments based on factors like risk, return and diversification operate by choosing options that provide the most value.

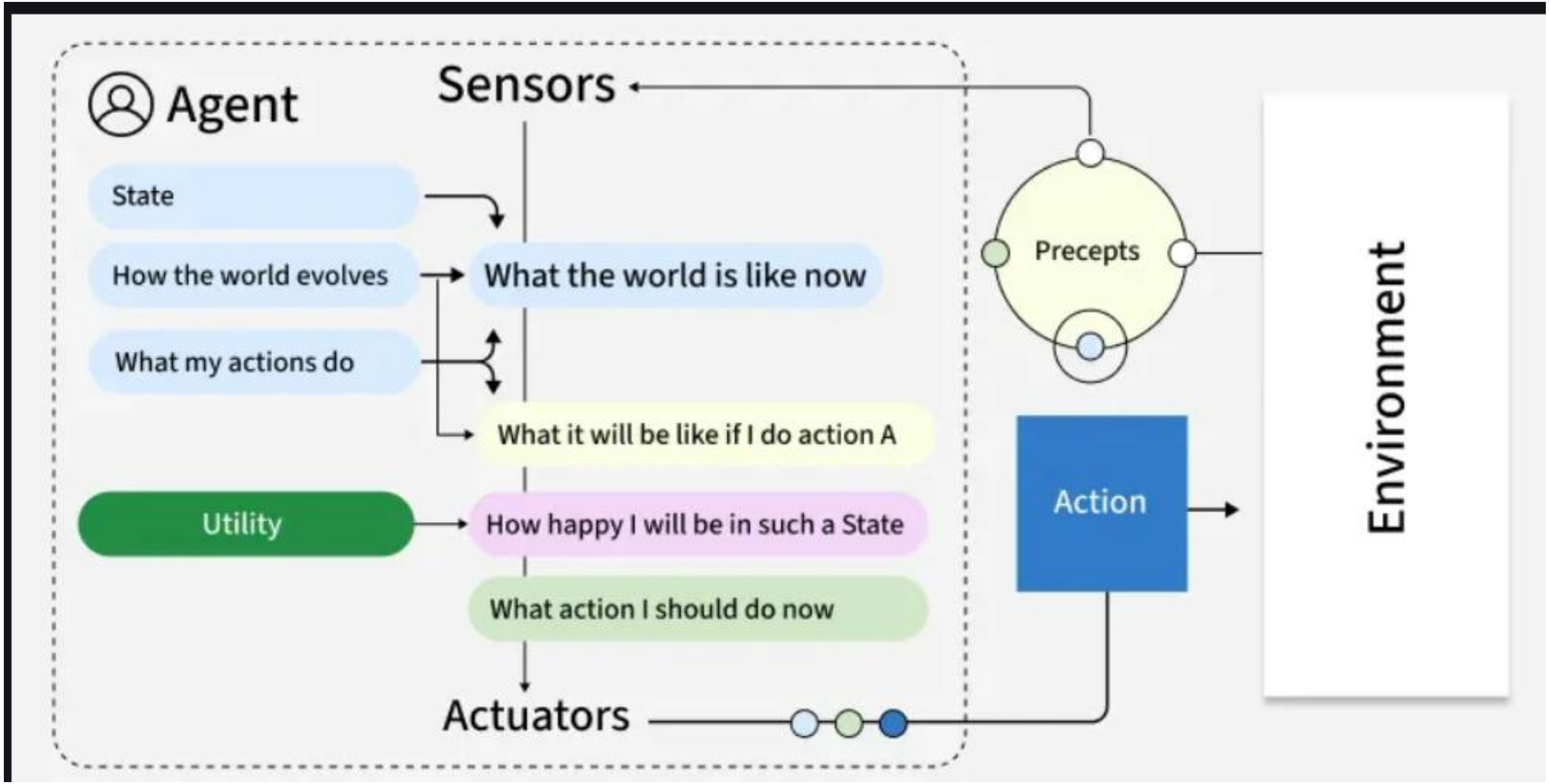


Figure: Utility-Based Agents

THANK You